Structured & Cybernetic Models of Microbial Growth

Going beyond Monod: giving the cell an internal state and a strategy

Biochemical Engineering · Based on Villadsen, Nielsen & Lidén, Bioreaction Engineering Principles (3rd ed.), Sect. 7.4, and the cybernetic-modelling literature of Ramkrishna and co-workers

Why unstructured models are not enough

Monod-type kinetics describe balanced growth only — the cell is a single variable, biomass X

Unstructured (Ch. 7.3)

- Biomass is one lumped quantity; μ = f(s) responds instantly to the environment
- Implies a fixed cell composition, so it can only be valid for balanced growth
- Cannot describe: lag phases, shift-up / shift-down transients, diauxic growth, changing RNA/protein content with μ

Structured (Ch. 7.4)

- Biomass split into ≥ 2 components (compartments, key enzymes, cybernetic variables)
- Intracellular state carries memory of the past → transients emerge naturally
- 'Simple' structure: a few well-chosen variables, not a full metabolic network

Model classification (after Tsuchiya, Fredrickson & Aris)

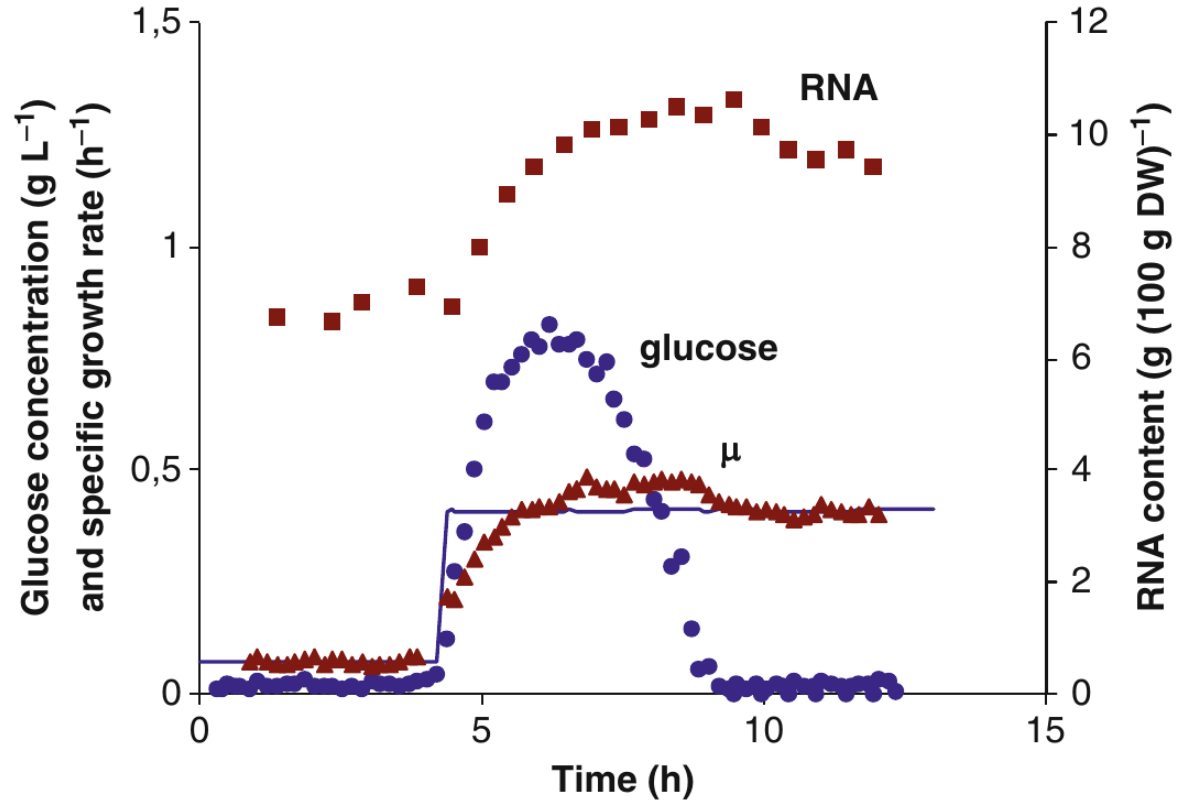
Unstructured, non-segregated Monod, Contois — the workhorse	**Structured, non-segregated** Compartment & cybernetic models ← this lecture
Unstructured, segregated Cell-size / age distributions	**Structured, segregated** Population balances with internal state

Rows: cell composition lumped vs. resolved. Columns: population treated as average cell vs. distribution of individuals.

Villadsen, Nielsen & Lidén (2011), Ch. 7.1–7.3; Fredrickson (1976) Biotechnol. Bioeng. 18:1481

Background I: the cell's machinery follows μ

Fig. 7.12 — RNA content and μ during a shift-up in a glucose-limited chemostat (Lactococcus cremoris)



Symbols: ● glucose (g L^{-1}), ▲ μ (h^{-1}), ■ RNA (g per 100 g DW, right axis); line = dilution rate D

The experiment

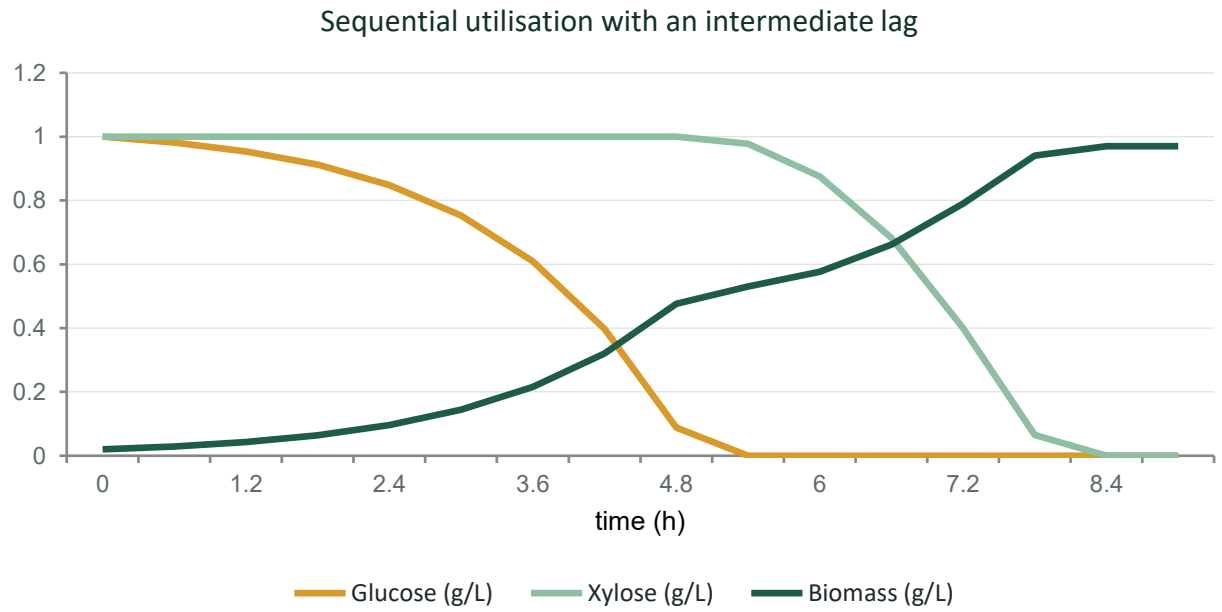
- Steady state at D = 0.075 h^{-1}; at t = 4.3 h, D is stepped to 0.409 h^{-1}
- Glucose makes a large excursion: ≈ 5.5 mg L^{-1} → 0.75 g L^{-1} → 10 mg L^{-1}
- μ climbs over 2–3 h and overshoots before settling at the new D
- RNA (≈ 80 % ribosomal) rises from ≈ 7 to ≈ 10 g per 100 g DW and mirrors μ

Why it matters

- $X_{RNA} \propto \mu$ at steady state for many organisms (Fig. 7.11) — and even through the transient
- The protein-synthesizing system (PSS) is therefore a natural "active" compartment
- Monod kinetics would let μ jump instantly with s; it cannot produce this lag

Fig. 7.12 reproduced from Villadsen, Nielsen & Lidén (2011), p. 302; data from Benthin et al. (1991)

Worked case: diauxic growth on two sugars



Structure after Kompala et al. (1984, 1986); parameters chosen for illustration, not fitted. Experimental validation: K. oxytoca on glucose/xylose/arabinose mixtures.

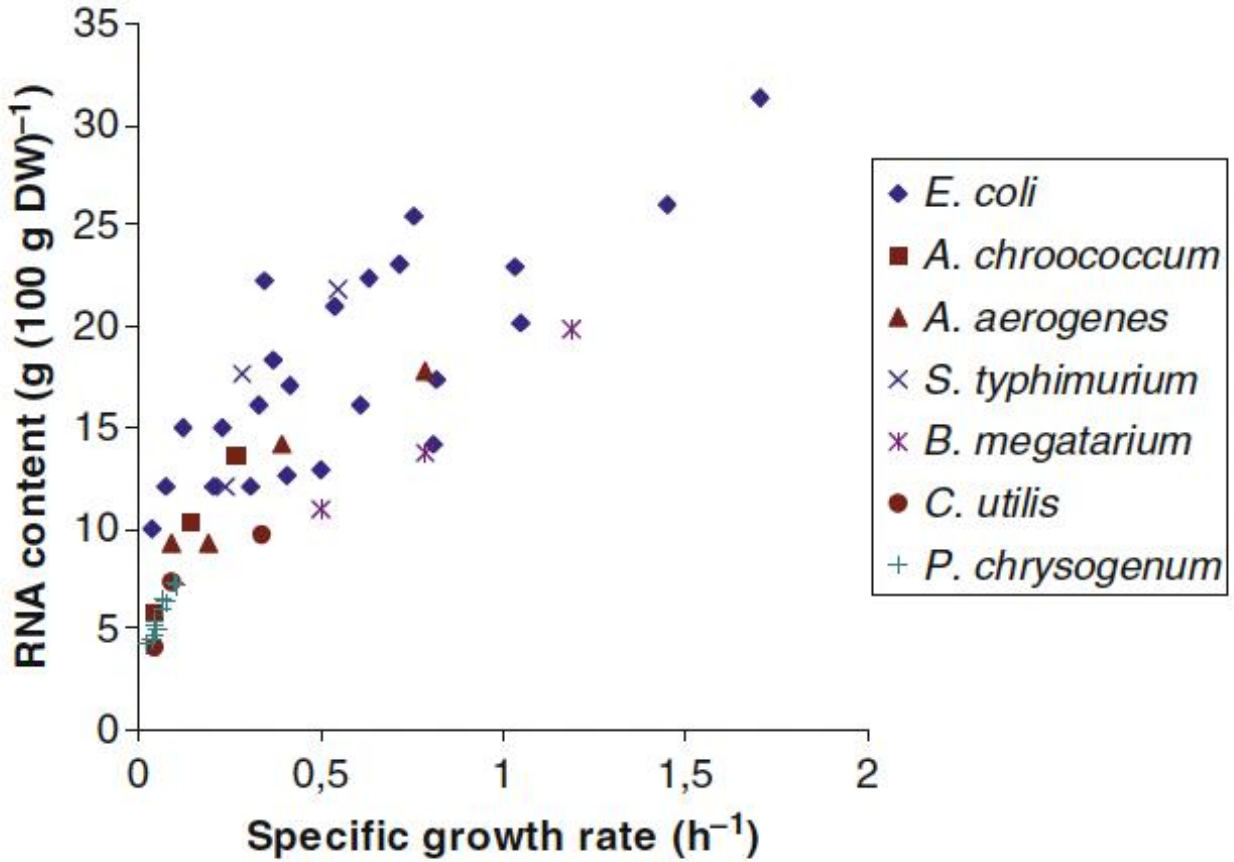
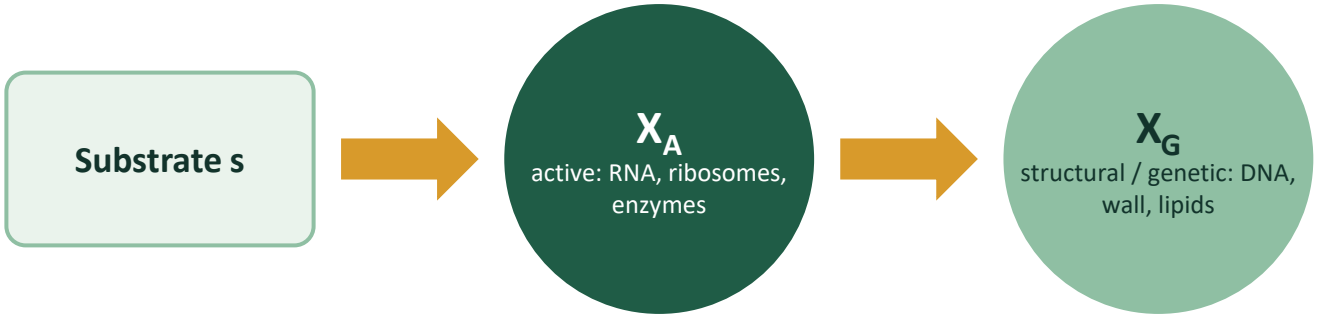


Fig. 7.11 Measurements of steady-state RNA content in various microorganisms as a function of the specific growth rate (equal to the dilution rate). The data are compiled from various literature sources

Compartment models: the two-compartment idea

Williams (1967) — the first structured model; Nielsen et al. (1991) — the version in the textbook



$$\text{Reaction 1:}\quad S \rightarrow X_{\mathrm{A}}, \qquad r_1 = k_1 \frac{s}{K_s + s}\ \text{(Monod in } s\text{)}$$

$$\text{Reaction 2:}\quad X_{\mathrm{A}} \rightarrow X_{\mathrm{G}}, \qquad r_2 \text{ first order in } X_{\mathrm{A}}$$

Key modelling choices

- Biomass = $X_A + X_G$, with mass fractions $x_A + x_G = 1$
- Only the active compartment X_A synthesises new material — it is the 'growth machinery'
- Rates are written per unit biomass; a dilution term $-\mu \cdot x_A$ appears because the cell volume grows
- Two ODEs for the fractions replace one for X; parameters stay few (k, K_s, Y)
- Any structured model must reduce to a Monod-like law at balanced growth — a useful consistency check

Williams (1967) J. Theor. Biol. 15:190; Nielsen, Nikolajsen & Villadsen (1991) Biotechnol. Bioeng. 38:1–10; VNL Sect. 7.4.1

The bookkeeping behind every structured model

VNL Sect. 7.2: write the reactions, assign rates — growth rate and balances then follow automatically

Specific growth rate — sum over the K biomass components

$$\mu = r_x = \sum_{k=1}^{K} \boldsymbol{\Gamma}_k^{\mathsf{T}}\, \mathbf{v} \tag{7.11}$$

Balance for component k (mass fraction X_k, g g^{-1} DW)

$$\frac{dX_k}{dt} = \boldsymbol{\Gamma}_k^{\mathsf{T}}\, \mathbf{v} \;-\; \mu X_k, \qquad \sum_{k=1}^{K} X_k = 1 \tag{7.14}$$

Reactor balances (chemostat, sterile feed; batch: D = 0)

$$\frac{dx}{dt} = (\mu - D)\, x, \qquad \frac{ds}{dt} = D\,(s_f - s) - q_s\, x$$

1 What Γ is

Rows = reactions j, columns = biomass components k. Γ_k is column k: how much of X_k each reaction makes (+) or uses (−).

2 Dilution term

$-\mu X_k$: as the cell grows, every component is diluted. To keep its fraction constant it must be made at rate μX_k. Valid in any reactor.

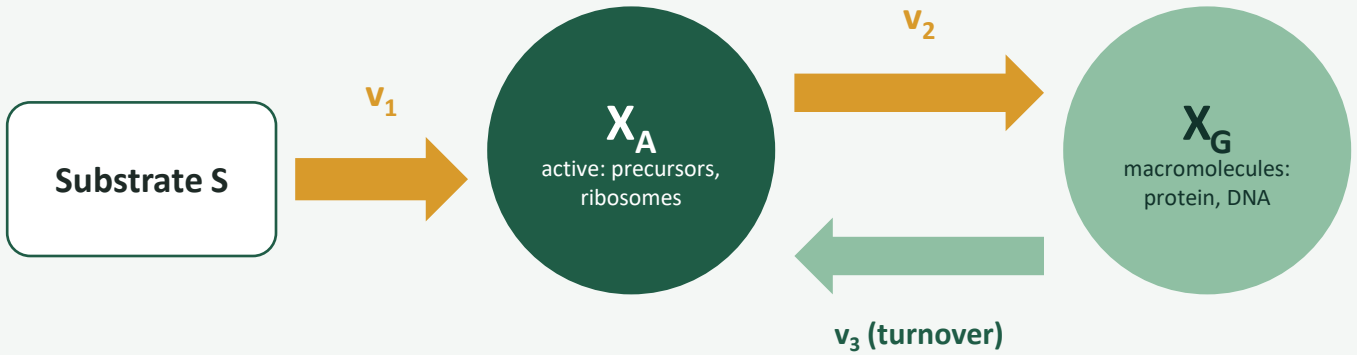
3 Consistency check

Summing (7.14) over k with $\Sigma X_k = 1$ returns (7.11). Reactions that only move mass between compartments add nothing to μ.

Villadsen, Nielsen & Lidén (2011), eqs. (7.11)–(7.14); used below for the Williams, LAB and cybernetic models

Williams model (1967): reactions and rates

Two compartments, three reactions — as corrected by Fredrickson (1976); VNL eqs. (7.32a)–(7.32c)



Why these rate forms?

- v_1: Monod uptake of S into the precursor pool
- $v_2 \propto X_A X_G$: part of G holds the enzymes that build G
- v_3: first-order decay of G refills A

$$\gamma_{11} X_\mathrm{A} - S = 0, \qquad v_1 = k_1 \frac{s}{s + K_s} \tag{7.32a}$$

S → A: synthesis of active material

$$\gamma_{22} X_\mathrm{G} - X_\mathrm{A} = 0, \qquad v_2 = k_2 \, X_\mathrm{A} X_\mathrm{G} \tag{7.32b}$$

A → G: macromolecule synthesis

$$X_\mathrm{A} - X_\mathrm{G} = 0, \qquad v_3 = k_3 \, X_\mathrm{G} \tag{7.32c}$$

G → A: turnover / recycling

Williams (1967) J. Theor. Biol. 15:190; Fredrickson (1976) Biotechnol. Bioeng. 18:1481; VNL Sect. 7.4.1. γ = stoichiometric (yield-type) coefficients; X in g (g DW)$^{-1}$

Williams model: growth rate, balances, steady state

Worked through in VNL Example 7.5

Stoichiometric matrices (rows = reactions 1–3; Γ columns = X_A, X_G)

$$\mathbf{A} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{\Gamma} = \begin{pmatrix} \gamma_{11} & 0 \\ -1 & \gamma_{22} \\ 1 & -1 \end{pmatrix}$$

Specific growth rate from (7.11)

$$\begin{aligned} \mu &= \gamma_{11} v_1 - (1 - \gamma_{22})\, v_2 \\ &= \gamma_{11} k_1 \frac{s}{s + K_s} - (1 - \gamma_{22})\, k_2 X_\mathrm{A} X_\mathrm{G} \end{aligned}$$

Reaction 3 only recycles biomass, so it adds nothing to μ.

Route to the steady state

$$\frac{dX_\mathrm{G}}{dt} = 0, \quad X_\mathrm{G} \neq 0 \;\Rightarrow\; \gamma_{22} k_2 X_\mathrm{A} = k_3 + \mu$$

Compartment balances from (7.14)

$$\frac{dX_\mathrm{A}}{dt} = \gamma_{11} k_1 \frac{s}{s + K_s} - k_2 X_\mathrm{A} X_\mathrm{G} + k_3 X_\mathrm{G} - \mu X_\mathrm{A}$$

$$\frac{dX_\mathrm{G}}{dt} = \gamma_{22} k_2 X_\mathrm{A} X_\mathrm{G} - k_3 X_\mathrm{G} - \mu X_\mathrm{G}$$

with $X_A + X_G = 1$. The X_A balance is eq. (4) of Ex. 7.5; the X_G balance follows the same way.

Steady-state chemostat (μ = D)

$$X_\mathrm{A} = \frac{D + k_3}{\gamma_{22}\, k_2}$$

- ✓ X_A rises linearly with D, as in Fig. 7.11
- ✗ fitted $k_3 \approx 0.5\ \mathrm{h}^{-1}$ — far too high
- ✗ $k_3 = 0$ gives $X_A = 0$ at D = 0

VNL Example 7.5. Despite its flaws the model fits fed-batch K. pneumoniae better than Monod + maintenance (Esener et al. 1981–82) and Z. mobilis $q_s(D)$ data (Jöbses et al. 1985)

LAB model: active and genetic compartments

Nielsen, Nikolajsen & Villadsen (1991) for Lactococcus cremoris — VNL Example 7.6, glucose as the limiting substrate

X_A

Active compartment

PSS (ribosomes) + small building blocks

X_G

Structural & genetic

All other cell material: protein, DNA, envelope

Reading the coefficients

- γ_{11}: a yield — lactate, CO_2 are left out of the stoichiometry
- γ_{22} slightly < 1: by-products formed when A → G

Stoichiometry

$$\gamma_{11} X_{\mathrm{A}} - S = 0, \qquad \gamma_{22} X_{\mathrm{G}} - X_{\mathrm{A}} = 0$$

(1), (2)

Kinetics: macromolecule synthesis is rate-controlling and catalysed by X_A

$$v_i = k_i \frac{s}{s + K_{s,i}} X_{\mathrm{A}}, \qquad i = 1, 2$$

(3)

Specific growth rate from (7.11)

$$\mu = \gamma_{11} v_1 - (1 - \gamma_{22})\, v_2 = \left(\gamma_{11} k_1 \frac{s}{s + K_{s,1}} - (1 - \gamma_{22})\, k_2 \frac{s}{s + K_{s,2}} \right) X_{\mathrm{A}}$$

(4), (5)

μ is proportional to X_A. Substrate acts twice: directly through $s/(s+K_{s,i})$, and indirectly because s sets the size of X_A.

Nielsen, Nikolajsen & Villadsen (1991) Biotechnol. Bioeng. 38:1–10 and 11–23; VNL Example 7.6 eqs. (1)–(5)

What structure buys you

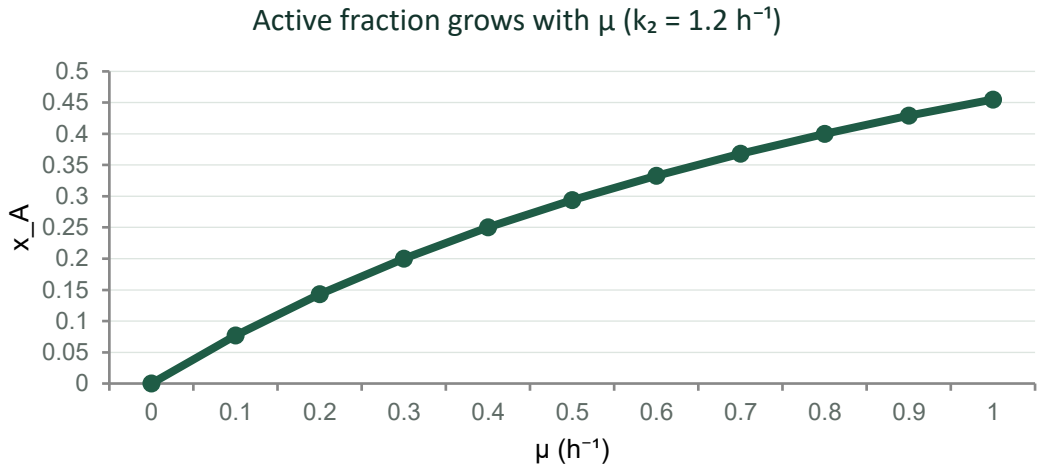
Simplest case — the LAB model with $\gamma_{11} = \gamma_{22} = 1$ and $s \gg K_{s,2}$: balance for X_A and its consequences

Balance on the active compartment (per unit biomass)

$$\frac{dX_\mathrm{A}}{dt} = r_1 - r_2 - \mu X_\mathrm{A}$$

$$r_1 = k_1 \frac{s}{K_1 + s} X_\mathrm{A}, \quad r_2 = k_2 X_\mathrm{A}, \quad \mu = r_1$$

$$\text{Balanced growth: } X_\mathrm{A} = \frac{\mu}{\mu + k_2} \Rightarrow \mu = k_1 \frac{s}{K_1 + s} - k_2$$



1 Composition vs. μ

RNA / ribosome content rises roughly linearly with growth rate — matches chemostat data for E. coli, yeast, and many others.

2 Lag phase

Cells from a slow-growth state have little X_A; after a shift-up the machinery must first be built, so growth lags even though s is high.

3 Shift-down / washout

After a drop in s, X_A is 'too large' and is diluted only slowly — the model overshoots and then relaxes, as observed.

Form of the balance and the μ–x_A relation follow VNL Sect. 7.4.1; chart is illustrative (k_2 chosen for display)

LAB model: balances, steady state and yield

What the two-compartment structure predicts

Compartment balances from (7.14)

$$\frac{dX_\mathrm{A}}{dt} = \gamma_{11} v_1 - v_2 - \mu X_\mathrm{A}$$

$$\frac{dX_\mathrm{G}}{dt} = \gamma_{22} v_2 - \mu X_\mathrm{G}$$

Steady-state chemostat ($\mu = D$, $X_G = 1 - X_A$)

$$X_\mathrm{A} = \frac{D}{D + \gamma_{22}\, k_2 \dfrac{s}{s + K_{s,2}}}$$

From the X_G balance. X_A increases with D and tends to 0 as $D \to 0$; with $\gamma_{22} = 1$ and $s \gg K_{s,2}$ it is $X_A = \mu/(\mu + k_2)$ (next slide).

Which compartment does high s favour?

$$\frac{v_1}{v_2} = \frac{k_1}{k_2} \frac{s + K_{s,2}}{s + K_{s,1}} \qquad (6)$$

If $K_{s,1} > K_{s,2}$, synthesis of X_A gains at high s — so the active fraction grows with μ.

Yield without any maintenance term

$$Y_{sx} = \frac{\mu}{v_1} = \gamma_{11} - (1 - \gamma_{22}) \frac{k_2}{k_1} \frac{s + K_{s,1}}{s + K_{s,2}}$$

The last term shrinks as s (and μ) rise, so Y_{sx} falls at low μ — the model mimics maintenance with no extra parameter.

VNL Example 7.6 eqs. (6)–(7). Y_{sx} written directly as μ/v_1 from eqs. (1)–(5); X_A steady state derived from the X_G balance

Background II: the culture remembers its history

Fig. 7.13 — wash-out of L. cremoris after D is stepped to 0.99 h^{-1}, above the critical D ≈ 0.55 h^{-1}

Monod prediction

After the step $s \gg K_s$ in both runs, so $\mu = \mu_{max}$ at once:

$$\ln \frac{x}{x_0} = (\mu_{\max} - D)\, t \quad \Rightarrow \quad \text{identical curves}$$

Two-compartment prediction

$$\mu(t) \simeq \left(\gamma_{11} k_1 - (1 - \gamma_{22}) k_2\right) X_{\mathrm{A}}(t)$$

- Fast: $s/(s + K_{s,i})$ jumps to ≈ 1 within minutes
- Slow: X_A must be built before μ can reach μ_{max}
- Cells from D = 0.10 h^{-1} start with little X_A → grow slowly → wash out faster
- Lines use parameters from steady-state data only

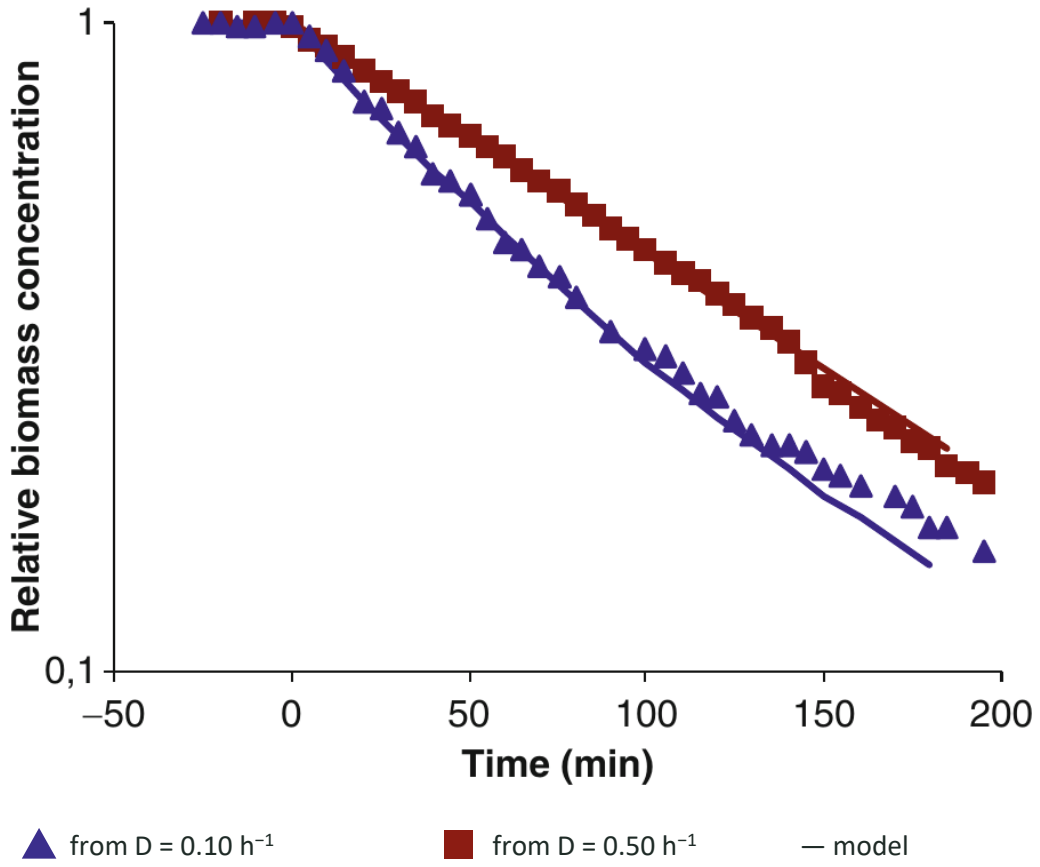


Fig. 7.13 reproduced from Villadsen, Nielsen & Lidén (2011), p. 307; data Nielsen et al. (1991b). Biomass normalised by its steady-state value before the step

Cybernetic models: the cell as an optimal strategist

Ramkrishna & co-workers (Purdue), from Kompala et al. 1984 onwards

The core idea

Instead of modelling the regulatory mechanism (repressors, cAMP, inducers …), assume the cell allocates scarce resources so as to maximise an objective — typically its instantaneous growth rate. Regulation is then the solution of an optimisation problem, not a set of extra kinetic terms.

Why bother?

- Mechanistic regulation is rarely known well enough to write kinetics for
- Every new inhibition/induction term adds parameters
- Optimality laws add almost no parameters and stay stable across phenotypes

u_i

Enzyme synthesis

Fraction of the ‘induction budget’ spent on enzyme E_i. Matching law:

$$u_i = \frac{r_i^*}{\sum_j r_j^*}$$

w_i

Enzyme activity

Degree to which existing E_i is switched on. Proportional law:

$$w_i = \frac{r_i^*}{\max_j r_j^*}$$

r_i^* = potential rate on substrate i alone (Monod kinetics × enzyme level). Only u_i and w_i are new — they add no parameters.

Kompala, Ramkrishna & Tsao (1984) Biotechnol. Bioeng. 26:1272; Dhurjati et al. (1985); Ramkrishna & Song (2012) AIChE J. 58:986; VNL Sect. 7.4.2

Mixed substrates: what a model has to reproduce

Growth on truly substitutable substrates — after Ramkrishna et al. (1987), VNL Table 7.6

1	**Preference**	The substrate that supports the fastest growth is used first → sequential uptake (diauxie) in batch.
2	**Conditional mixing**	Uptake becomes simultaneous when another nutrient (e.g. the N source) is limiting.
3	**No synergy**	Even with simultaneous uptake, μ never exceeds the best single-substrate value.
4	**Fast switch-off**	Adding a “faster” substrate during growth on a “slower” one quickly stops use of the slower one.
5	**Chemostat**	Mixed uptake at low D; the faster substrate is taken preferentially at high D.

The key-enzyme idea

Growth on S_i needs a bottleneck enzyme E_i — in reality often several lumped together (e.g. lactose permease + β-galactosidase).

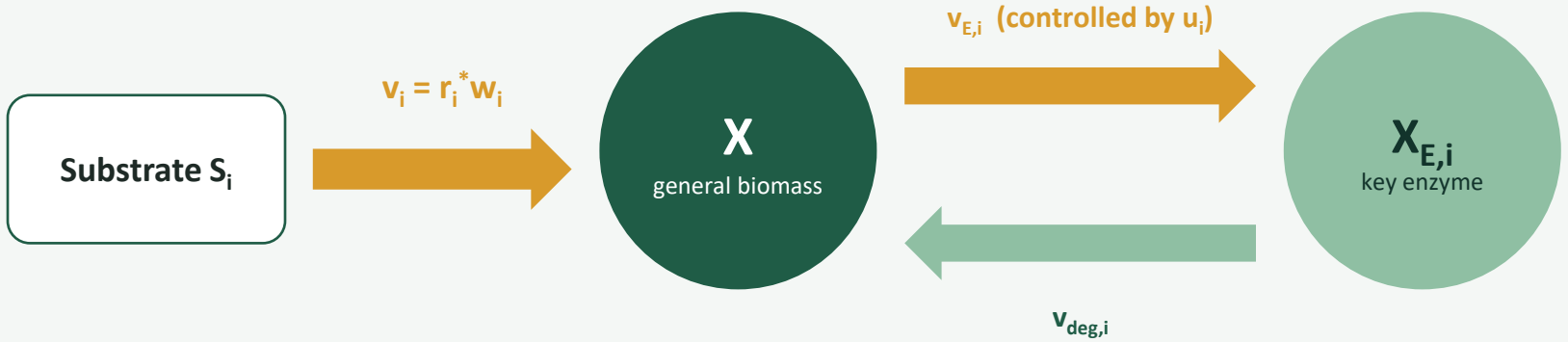
E_i must be synthesised before S_i can be used: the diauxic lag is the time to build it.

Everyday example: in bread dough, yeast makes CO_2 first from glucose, then from maltose.

Ramkrishna, Kompala & Tsao (1987); VNL Sect. 7.4.2 and Table 7.6 (paraphrased)

Cybernetic model: three reactions per substrate

Kompala et al. (1984, 1986) — VNL eqs. (7.33)–(7.35), for substrates i = 1 … N



$X_{E,i}$ catalyses uptake of S_i.

r_i^* = potential uptake rate: Monod in s_i × enzyme level, with no cybernetic throttling.

$$\gamma_i X - S_i = 0, \qquad v_i = r_i^* w_i = k_i \frac{s_i}{s_i + K_i} X_{E_i} w_i \qquad (7.33)$$

growth on S_i

$$X_{E_i} - X = 0, \qquad v_{E,i} = k_{E,i} \frac{s_i}{s_i + K_i^E} u_i \qquad (7.34)$$

inducible synthesis of E_i

$$X - X_{E_i} = 0, \qquad v_{\mathrm{deg},i} = k_{\mathrm{deg},i} X_{E_i} \qquad (7.35)$$

first-order enzyme decay

Kompala, Ramkrishna & Tsao (1984) Biotechnol. Bioeng. 26:1272; Kompala et al. (1986) Biotechnol. Bioeng. 28:1044; VNL Sect. 7.4.2

The cybernetic variables u_i and w_i

Resource allocation laws chosen by Kompala et al. (1986) — VNL eqs. (7.36)–(7.37)

u_i Enzyme synthesis — matching law

$$u_i = \frac{r_i^*}{\sum_{j=1}^{N} r_j^*} \qquad (7.36)$$

Synthesis resources are shared in proportion to the return; $\Sigma u_i = 1$. Control at the level of gene expression.

w_i Enzyme activity — proportional law

$$w_i = \frac{r_i^*}{\max_j \left(r_j^*\right)} \qquad (7.37)$$

The best substrate runs at full activity (w = 1); the others are throttled. Control at the level of the enzyme.

Two quick cases

Situation	r_1^*, r_2^*	u_1, u_2	w_1, w_2	Consequence
E_2 not yet induced	1.00, 0.02	0.98, 0.02	1.00, 0.02	Grows on S_1; almost no E_2 made → sequential use, diauxic lag
Two equally good substrates	0.50, 0.50	0.50, 0.50	1.00, 1.00	Both used at full rate. A matching law for w would halve μ — why (7.37) was preferred

VNL eqs. (7.36)–(7.37); Kompala et al. (1986). Rates r^* in the table are illustrative (same units). Young & Ramkrishna (2007) later derived both laws from an optimal-control problem

Cybernetic model: the complete set of equations

Kompala model with constitutive enzyme synthesis (Turner & Ramkrishna 1988) — batch or chemostat

Substrate uptake

$$v_i = k_i \frac{s_i}{s_i + K_i} X_{E_i} w_i \qquad (7.33)$$

Specific growth rate

$$\mu = \sum_{i=1}^{N} \gamma_i v_i \qquad (7.11)$$

Key enzyme

$$\frac{dX_{E_i}}{dt} = k_{E,i} \frac{s_i}{s_i + K_i^E} u_i - k_{\mathrm{deg},i} X_{E_i} + k_{\mathrm{con},i} - \mu X_{E_i} \qquad (7.38)$$

Substrates

$$\frac{ds_i}{dt} = D\left(s_{f,i} - s_i\right) - v_i x$$

Biomass

$$\frac{dx}{dt} = (\mu - D)\, x \qquad (\text{batch: } D = 0)$$

Control laws

$$u_i = \frac{r_i^*}{\sum_j r_j^*}, \qquad w_i = \frac{r_i^*}{\max_j r_j^*}, \qquad r_i^* = k_i \frac{s_i}{s_i + K_i} X_{E_i} \qquad (7.36\text{–}37)$$

Why $\mu = \Sigma \gamma_i v_i$

Enzyme synthesis and decay only move mass between X and $X_{E,i}$; they make no new biomass.

Why $k_{con,i}$

Without it, a chemostat fed one sugar loses every other enzyme and could never take up a pulse of another sugar — contradicted by experiment.

Parameters

Per substrate: k_i, K_i, γ_i, $k_{E,i}$, K_i^E, $k_{deg,i}$, $k_{con,i}$. All from single-substrate runs; mixed-substrate runs are then predictions.

VNL eqs. (7.33)–(7.38); μ from (7.11). Reactor balances as on the bookkeeping slide. Kompala et al. (1986) reproduced diauxic and triauxic batches with this structure

Worked case: diauxic growth on two sugars

Kompala-type model, two substrates, two inducible enzymes — simulated

Model equations (i = 1, 2) in textbook notation

$$r_i^* = k_i \frac{s_i}{K_i + s_i} X_{E_i}, \qquad \mu = \sum_i \gamma_i r_i^* w_i$$

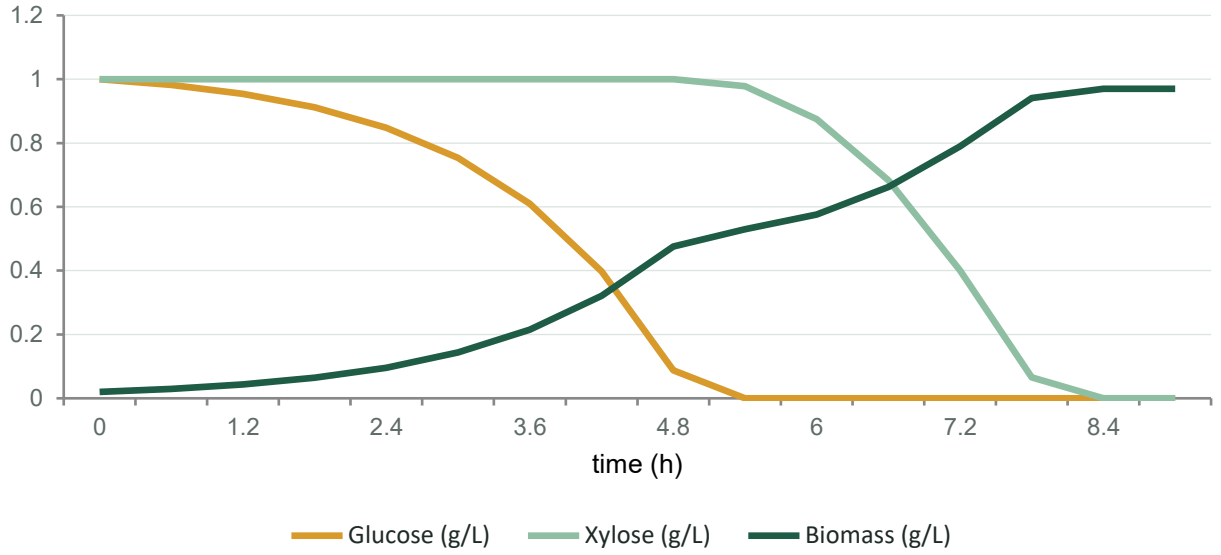
$$\frac{ds_i}{dt} = -r_i^* w_i x, \qquad \frac{dx}{dt} = \mu x$$

$$\frac{dX_{E_i}}{dt} = k_{E,i} \frac{s_i}{K_i^E + s_i} u_i + k_{\mathrm{con},i} - (k_{\mathrm{deg},i} + \mu) X_{E_i}$$

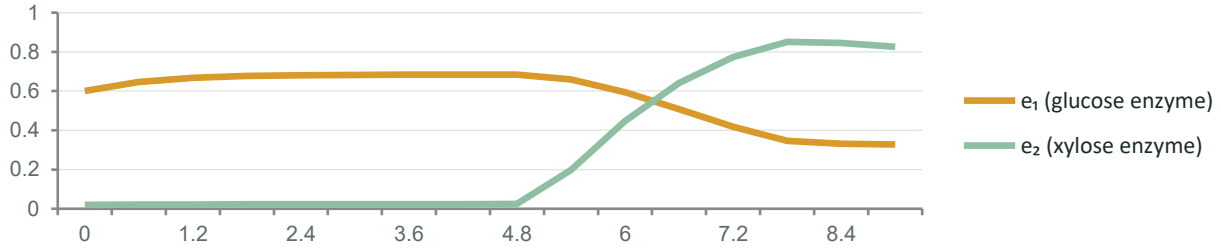
$$u_i = \frac{r_i^*}{\sum_j r_j^*}, \qquad w_i = \frac{r_i^*}{\max_j r_j^*}$$

$X_{E,i}$ = relative enzyme level (e_i in the lower chart); $k_{con,i}$ = small constitutive synthesis; $k_{deg,i}$ = enzyme decay. Illustrative values: $\gamma_i k_i$ = 1.0 / 0.45 h^{-1}, K_i = 0.01 / 0.05 g L^{-1}, γ_i = 0.5 / 0.45.

Sequential utilisation with an intermediate lag



Enzyme levels: e_2 is only built once glucose is gone ($u_2 \approx 0$ before)



Structure after Kompala et al. (1984, 1986); parameters chosen for illustration, not fitted. Experimental validation: K. oxytoca on glucose/xylose/arabinose mixtures.

Compartment vs. cybernetic, and what came next

	Compartment models	Cybernetic models
What is 'structured'	Biomass split into 2–3 pools (active / structural)	Key enzymes + control variables u_i, w_i
Regulation	Implicit — rates depend on pool sizes	Explicit — optimal allocation of synthesis & activity
Extra parameters	A few rate constants	Enzyme synthesis/decay (k_E, k_{deg}); u, w add none
Best at	Lag, shift-up/down, composition vs μ	Mixed substrates, diauxie, switching between pathways
Weakness	Compartments are abstractions, hard to measure	Objective function is a hypothesis; can miss non-optimal behaviour

Cybernetic modelling since the textbook

2007 — Matching & proportional laws shown to be the solution of an optimal-control problem (Young & Ramkrishna, Biotechnol. Prog.)

2008–11 — Hybrid cybernetic models: control laws applied to elementary flux modes of a metabolic network; lumped HCM predicts knockout strains from wild-type data (Song & Ramkrishna)

2012 — Review: 'Dynamic models of metabolism — the cybernetic approach' (AIChE J.)

2020s — Reinterpreted via LP / resource allocation; links to proteome-allocation models of growth

Young & Ramkrishna (2007) Biotechnol. Prog. 23:83; Song & Ramkrishna (2010) Biotechnol. Bioeng. 106:271; Ramkrishna & Song (2012) AIChE J. 58:986

Take-aways

1 Structure = memory

Giving the cell internal variables lets transients (lag, shift, diauxie) emerge instead of being bolted on.

2 Keep it simple

Two compartments or two control variables already capture most of the qualitative behaviour with few parameters.

3 Optimality as a modelling tool

Cybernetic laws replace unknown regulatory mechanisms by a hypothesis about what the cell is trying to do — testable and cheap.

4 Check the limit

Every structured model should collapse to a Monod-like law under balanced growth.

Discussion question: Monod's E. coli grow on glucose + lactose. Which model — compartment or cybernetic — would you use to predict the length of the diauxic lag, and what would you need to measure to fit it?